

Physics, 2017 EC, entrance exam, version 2

Grade 12 · Physics · 2017 E.C.

Time allowed: 120 minutes

Questions: 50

Total marks: 50

Name _____ Grade _____ ID No. _____ Date _____

1. Two trains are traveling towards each other on parallel tracks. Train X is moving west at 80 km/h, Train Y moving east at 60 km/h. If you are on Train X, what is the velocity of Train Y relative to Train X? (1)
 - A. 140 km/h towards the east
 - B. 60 km/h towards the west
 - C. 20 km/h towards the west
 - D. 140 km/h towards the west
2. Which of the following is a correct action-reaction force pair? (1)
 - A. The force of a ball hitting a wall and the force of gravity pulling the ball down.
 - B. The force of a car's tires pushing backward on the road and the road pushing the car forward.
 - C. The force of the bat hitting a ball and the force of air resistance on the ball.
 - D. The force of a swimmer's arms pushing water backward and the force of gravity pulling them down.
3. A 0.25 kg ball moving at 4 m/s collides inelastically with a stationary ball of mass 0.75 kg. After collision they stick together. What is the kinetic energy lost? (1)
 - A. 4 J
 - B. 2 J
 - C. 3.5 J
 - D. 1.5 J
4. A block weighing 200 N is placed on a frictionless inclined plane that rises to a height of 3 m over a distance of 12 m along the plane. What is the mechanical advantage (MA) of the inclined plane? (1)
 - A. 1.67
 - B. 3.00
 - C. 4.00
 - D. 6.00
5. What is the main characteristic of a third-order lever? (1)
 - A. The fulcrum is at one end, with the effort on the other end.
 - B. The load is positioned between the effort and the fulcrum.
 - C. It has a mechanical advantage greater than one.
 - D. The effort is applied between the load and the fulcrum.
6. Considering the relationship between atmospheric pressure and altitude, which evaluation best explains the challenges faced by climbers at high elevations? (1)
 - A. Climbers are less affected by low atmospheric pressure because they are physically fit.
 - B. Decreased atmospheric pressure at high altitudes leads to reduced oxygen levels, which can impair physical performance and increase fatigue.
 - C. Climbers do not need to acclimatize as atmospheric pressure has no significant impact on their performance.
 - D. High atmospheric pressure at altitudes enhances climbers' endurance and reduces fatigue.
7. Which of the following is correct? (1)
 - A. Temperature is the measure of the average molecular kinetic energy of a gas
 - B. Internal energy of the system is independent of its mass or the number of molecules in the system
 - C. Heat is energy that can be transferred from one object to another at equal temperature
 - D. Temperature is a measure of the amount of energy that flows from hot to colder region

8. The sound intensity at a distance of 2 m from a point source is 1 W/m^2 . What will be the intensity at a distance of 4 m from the same source, assuming uniform spherical spreading? (1)

A. 0.25 W/m^2 B. 0.5 W/m^2
C. 1.0 W/m^2 D. 2 W/m^2

9. Which situation best shows the application of both experiential and non-experiential knowledge in solving a complex problem? (1)

A. A scientist uses experimental data from a lab experiment and theoretical models to refine their hypothesis.
B. A teacher applies established classroom management techniques without considering past experiences with students.
C. A lawyer follows court precedents and applies personal experiences from similar cases to argue in court.
D. A nurse consults a medical handbook, but only follows their instinct when making decisions.

10. How do physics graduates contribute to the healthcare sector? (1)

A. Designing promotional campaigns for health care products.
B. Working in pharmaceuticals for drug development.
C. Managing healthcare institutions.
D. Creating, reviewing and maintaining medical technologies in the field.

11. Which of the following defines a parallel vector? (1)

A. A vector that is parallel to the ground.
B. A vector that points in the same or exactly opposite direction to another vector.
C. A vector that has zero magnitude.
D. A vector that always points toward the center of the Earth.

12. The velocity of a particle moving with constant acceleration at an instant t_0 is 10 m/s . After 5 seconds, velocity is 20 m/s . The velocity at 3 seconds before t_0 is: (1)

A. 8 m/s B. 4 m/s
C. 6 m/s D. 7 m/s

13. A ball is dropped from a height of 80 m. How long will it take to hit the ground? ($g=10 \text{ m/s}^2$) (1)

A. 3.5 s B. 4.0 s
C. 5.0 s D. 6.0 s

14. A 5 kg box is resting on an incline of 30° . Coefficient of static friction = 0.58. What is the maximum static friction force before it slides? (1)

A. 25 N B. 15 N
C. 12.5 N D. 10 N

15. If the force applied to an object is perpendicular to its displacement, what is the work done? (1)

A. Maximum B. Negative
C. Half of the maximum D. Zero

16. Which of the following scenarios illustrates heat transfer by convection? (1)

A. A spoon becoming hot when placed in a hot bowl of food.
B. The sun warming the Earth through the vacuum of space.
C. A blanket keeping a person warm on a cold night.
D. A pot of water heating up from the bottom on a stove.

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17. Body A is ten times the mass and half the specific heat capacity of body B. If they receive equal heat, how do their heat capacities compare? (1)
- A. Heat capacity of A is 10 times greater
B. Heat capacity of B is 10 times greater
C. Heat capacity of A is 5 times greater
D. Heat capacity of B is 5 times greater
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18. Two rods of length L and $2L$ with coefficients of linear expansion α and 2α respectively are connected so total length becomes $3L$. The average coefficient of linear expansion of the composite rod equals: (1)
- A. $(3/2)\alpha$
B. $(5/3)\alpha$
C. $(5/2)\alpha$
D. α
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19. When ice melts into water at 0°C , what happens to the temperature of the system until all ice has converted? (1)
- A. Temperature increases continuously until 100°C .
B. Temperature decreases until all ice has melted.
C. Temperature remains constant at 0°C despite heat addition.
D. Temperature fluctuates between 0°C and 5°C .
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20. Two small spheres with charges $q_1=3\text{ }\mu\text{C}$ and $q_2=-2\text{ }\mu\text{C}$ are placed 6 cm apart in vacuum. What is the magnitude of electrostatic force between them? ($k=9\times 10^9\text{ Nm}^2/\text{C}^2$) (1)
- A. 1.5 N
B. 9 N
C. 3 N
D. 15 N
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21. A chef is preparing food in a clay pot and iron pot of the same size on identical stoves. The food in clay pot takes longer to heat but stays warm longer. Best explanation? (1)
- A. The clay pot has higher specific heat capacity, requiring more heat to raise its temperature but retaining it longer.
B. The iron pot has a lower specific heat capacity, so it transfers heat to the food more slowly.
C. The clay pot reflects heat, preventing rapid heating but ensuring gradual cooling.
D. The iron pot traps heat inside, causing the food to heat up faster but cool down quickly.
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22. Designing a circuit to convert a galvanometer (full-scale 5 mA, resistance $30\text{ }\Omega$) into a voltmeter measuring up to 50 V. What series resistor is needed? (1)
- A. $9,970\text{ }\Omega$
B. $10,000\text{ }\Omega$
C. $1,000\text{ }\Omega$
D. $2,000\text{ }\Omega$
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23. Which statement best describes why inserting a dielectric increases capacitance? (1)
- A. The dielectric reduces the distance between plates.
B. The dielectric allows higher voltage to be applied.
C. The dielectric polarizes and reduces the effective electric field between plates.
D. The dielectric increases the plate size, allowing more surface area.
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24. An ancient wooden artifact has only 25% of its original ^{14}C remaining. Half-life of ^{14}C is 5730 years. How old is the artifact? (1)
- A. 5,730 years
B. 11,460 years
C. 17,190 years
D. 22,920 years

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- 25.** What makes a body more stable according to Newtonian mechanics? (1)
- A. A narrower base support
 - B. The higher center of gravity
 - C. A wider base of support and a lower center of gravity
 - D. The material of the body
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- 26.** Which comparison accurately highlights a fundamental difference between nuclear fission and nuclear fusion? (1)
- A. Fission involves combining light nuclei, fusion involves splitting heavy nuclei.
 - B. Both require high temperatures, but only fission produces radioactive waste.
 - C. Fission occurs naturally in stars, fusion is used in nuclear reactors.
 - D. Fusion releases energy by combining light nuclei, whereas fission releases energy by splitting heavy nuclei.
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- 27.** Which statement best explains the relationship between physics and chemistry? (1)
- A. Physics is primarily concerned with matter, while chemistry focuses solely on energy.
 - B. Chemistry relies on the principles of physics to explain how atoms and molecules interact and behave.
 - C. Physics and chemistry are completely independent fields with no overlap.
 - D. Chemistry is a branch of physics that deals exclusively with chemical reactions.
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- 28.** Which scenario best illustrates the application of nuclear medicine in diagnosing a medical condition? (1)
- A. A patient undergoes an MRI scan to assess brain activity.
 - B. A physician uses a PET scan to visualize metabolic activity in a tumor.
 - C. A doctor prescribes antibiotics for a bacterial infection.
 - D. A patient receives a physical examination to check for signs of illness.
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- 29.** Which best explains how electrical properties of neurons facilitate communication? (1)
- A. Neurons use electrical signals only for muscle contraction, with no role in intercellular communication.
 - B. Electrical signals in neurons are uniform and do not encode different types of information.
 - C. Action potentials depend on selective permeability of neuronal membrane, allowing specific ion movements that encode information.
 - D. Neurons rely solely on chemical signals for communication, not electrical properties.
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- 30.** Which type of radiation, related to physics in medicine, is frequently used in cancer treatment? (1)
- A. Gamma-rays
 - B. Radio waves
 - C. Alpha particles
 - D. Microwaves
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- 31.** Which statement best evaluates the effectiveness of infrared wave detection compared to visible light in night vision systems? (1)
- A. Infrared detection is less effective because it cannot penetrate fog or smoke.
 - B. Infrared detection is superior because it detects heat emitted by objects, useful in low-light conditions.
 - C. Infrared detection provides a clearer image than visible light under all conditions.
 - D. Infrared detection is only useful in military applications, not civilian.
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- 32.** Which branch of physics is most crucial for understanding transmission of radio waves, microwaves, and infrared waves in communication technologies? (1)
- A. Thermodynamics
 - B. Quantum mechanics
 - C. Classical mechanics
 - D. Electromagnetism

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- 33.** At the peak of its trajectory, what is the vertical component of a projectile's velocity? (1)
- A. Maximum
B. Half of initial value
C. Zero
D. Constant
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- 34.** In projectile motion, which is the correct relation between total time of flight (T) and maximum height (H)? (1)
- A. $H = (1/4)gT^2$
B. $H = (1/2)gT^2$
C. $H = (1/8)gT^2$
D. $H = gT$
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- 35.** A fan blade starts from rest and reaches angular velocity 30 rad/s in 6 s with constant angular acceleration. What is angular displacement during this time? (1)
- A. 90 rad
B. 120 rad
C. 180 rad
D. 210 rad
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- 36.** A solid disk ($I = \frac{1}{2}mr^2$) and a hollow cylinder ($I = mr^2$) of same mass and radius roll down an incline without slipping. Which statement correctly compares their final velocities? (1)
- A. Solid disk has greater final velocity.
B. Hollow cylinder has greater final velocity.
C. Both have same final velocity.
D. Cannot be determined.
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- 37.** Rotating system: $m_1 = 2\text{kg}$ at $r_1 = 1\text{m}$, $m_2 = 3\text{kg}$ at $r_2 = 2\text{m}$, $m_3 = 4\text{kg}$ at $r_3 = 3\text{m}$. What is total moment of inertia I about the axis? (1)
- A. $34 \text{ kg}\cdot\text{m}^2$
B. $38 \text{ kg}\cdot\text{m}^2$
C. $42 \text{ kg}\cdot\text{m}^2$
D. $50 \text{ kg}\cdot\text{m}^2$
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- 38.** Which statement correctly summarizes all three of Kepler's Laws? (1)
- A. All planets have circular orbits; their speeds vary; their distances affect their speeds.
B. Planets orbit in straight lines, move at constant speeds, and have periods that depend on their masses.
C. Planets orbit in ellipses, sweep equal areas, and have periods related to their distances from the Sun.
D. All celestial bodies are attracted to each other; they move in random paths; and their speeds are equal.
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- 39.** Two moons orbit a planet. Moon A has orbital radius R, Moon B has radius 4R. How does orbital period of B compare to A? (1)
- A. Twice as long
B. 8 times as long
C. 4 times as long
D. 2 times as long
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- 40.** Two masses 4 kg and 6 kg separated by 2 m. Gravitational force between them? $G = 6.7 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$ (1)
- A. $4.0 \times 10^{-11} \text{ N}$
B. $8.0 \times 10^{-10} \text{ N}$
C. $2.0 \times 10^{-11} \text{ N}$
D. $4.0 \times 10^{-10} \text{ N}$
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- 41.** Kepler's third law is generalized by Newton by showing that it applies to: (1)
- A. Only planets orbiting the Sun
B. Any two orbiting bodies
C. Only elliptical orbits, not circular
D. Systems where gravity is negligible

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- 42.** What is the primary cause of pressure in a gas? (1)
- A. Gas particles colliding with the walls of their container
 - B. The weight of the gas
 - C. The volume of the gas
 - D. The temperature of the gas
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- 43.** Which correctly expresses the relationship between absolute pressure, gauge pressure, and atmospheric pressure? (1)
- A. Absolute pressure = gauge pressure minus atmospheric pressure.
 - B. Gauge pressure = absolute pressure plus atmospheric pressure.
 - C. Atmospheric pressure = absolute pressure plus gauge pressure.
 - D. Absolute pressure = gauge pressure plus atmospheric pressure.
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- 44.** What force is needed on a hydraulic press piston of radius 2 cm to produce 7200 N on working piston of radius 6 cm? (1)
- A. 2400 N
 - B. 800 N
 - C. 1050 N
 - D. 667 N
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- 45.** A scientist measures pressure at depth 1 m = 50 kPa. At depth 3 m in same liquid, what can she conclude? (1)
- A. Pressure increases as depth increases.
 - B. Pressure decreases as depth increases.
 - C. Pressure remains constant.
 - D. No consistent relationship.
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- 46.** A student uses an open-ended manometer. If mercury level is higher on gas side than open side, what is concluded? (1)
- A. Gas pressure is greater than atmospheric
 - B. Gas has no pressure
 - C. Gas pressure equals atmospheric
 - D. Gas pressure is less than atmospheric
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- 47.** A cube with side 0.5 m submerged in fluid density 1200 kg/m³. Buoyant force? (1)
- A. 1500 N
 - B. 500 N
 - C. 1200 N
 - D. 250 N
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- 48.** A liquid flows through horizontal pipe where diameter decreases from 10 cm to 5 cm. Velocity at larger section = 4 m/s. Velocity at smaller section? (1)
- A. 4 m/s
 - B. 8 m/s
 - C. 12 m/s
 - D. 16 m/s
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- 49.** Which best describes turbulent fluid flow? (1)
- A. Velocity is constant at every point
 - B. Motion is smooth and predictable
 - C. Fluid particles follow irregular, chaotic paths
 - D. Fluid flows with well-defined layers
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- 50.** During operation of a high-pressure steam turbine, you detect unusual noises and vibrations. What is your immediate response? (1)
- A. Increase steam flow to compensate.
 - B. Continue operation and report later.
 - C. Reduce load and initiate immediate inspection.
 - D. Temporarily shut down and wait for anomalies to resolve.

— END OF EXAMINATION —

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